

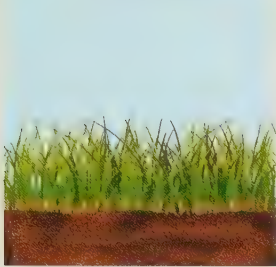
Figure 1-2

Salt Marsh Experiment

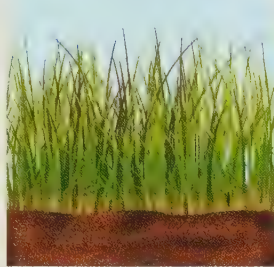
Salt marshes are coastal environments often found where rivers meet the sea. Researchers made an interesting observation about the way a particular species of marsh grass grows. They then applied scientific methodology to answer questions that arose from their observation.

1. OBSERVING AND ASKING QUESTIONS

Location A

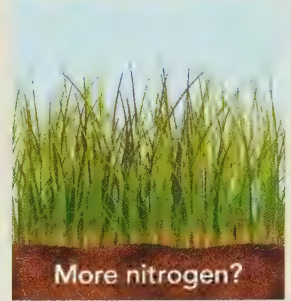
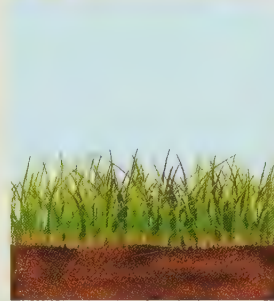


Location B



Researchers observed that marsh grass grows taller in some places than others. This observation led to a question: *Why do marsh grasses grow to different heights in different places?*

2. INFERRING AND HYPOTHESIZING



The researchers inferred that something limits grass growth in some places. It could be any environmental factor—temperature, sunlight, water, or nutrients. Based on their knowledge of salt marshes, they proposed a hypothesis: *Marsh grass growth is limited by available nitrogen.*

3. DESIGNING CONTROLLED EXPERIMENTS



The researchers selected similar plots of marsh grass. All plots had similar plant density, soil type, input of freshwater, and height above average tide level. The plots were divided into control and experimental groups.

Control Group

Experimental Group



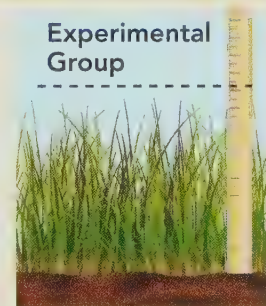
The researchers added nitrogen fertilizer (the independent variable) to the experimental plots. They then observed the growth of marsh grass (the dependent variable) in both experimental and control plots.

4. COLLECTING DATA

Control Group

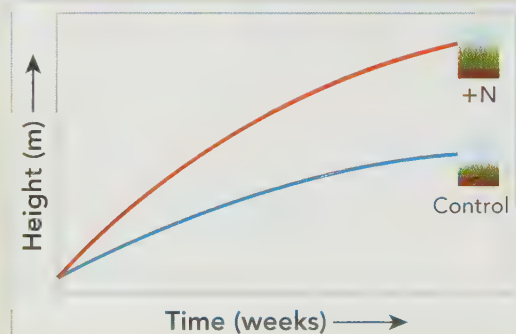


Experimental Group



The researchers sampled all the plots throughout the growing season. They measured growth rates and plant sizes, and analyzed the chemical composition of living leaves.

5. ANALYZING CONCLUSIONS



Data from all plots were compared and evaluated by statistical tests. Data analysis confirmed that marsh grasses in experimental plots with additional nitrogen did, in fact, grow taller and larger than controls. The hypothesis and its predictions were supported.